Final Project Proposal - Group 1

Supply Chain Pricing & Inventory Exchange Network

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Problem Statement:
Nowadays, many supply chains are facing the same issue from delayed communication, inaccurate inventory visibility, and slow replenishment decisions because each business manages its inventory independently.

Our project builds a centralized ecosystem that combines and connects a Manufacturer, a Wholesaler, and a Retailer through work requests, inventory exchange, pricing coordination, and production planning. The goal is to create a system where each enterprise controls its own operations and authority, but at the same time working across each other helps reduce stock shortages, minimize overstock/ unused stock, and speed up replenishment decisions across the network instead of operating independently.

Role covered:
Manufacturer: Production Analyst, Operations Manager
Wholesaler: Pricing Analyst, Marketing Specialist, and Sales Representative
Retailer: Retail Analyst, Store Manager, Store Associate

System Structure:
The system has one Network with three Enterprises, six Organizations, and eight roles (not counting admin roles):

Each enterprise's contribute to a total value where:

- Manufacturer - supplies production capacity and quotes. Without it, the network has no upstream source of goods or pricing baseline for new inventory.
- Wholesaler - bridges supply and demand: analyzes pricing, negotiates bulk terms with the Manufacturer, and markets overstock. Without it, retailers would negotiate production directly with manufacturers at far higher transaction cost.
- Retailer - captures real demand signals (store-level stock, sell-through) and turns them into restock and pricing decisions. Without it, the network has no visibility into actual consumer-facing inventory needs.

Work Request: At least six work request flows are implemented, with a minimum of two cross-organization and two cross-enterprise requested, as required.

- Production Analyst requests priority scheduling from Operations Manager (cross-organization)
- Pricing Analyst requests a manufacturing quote from Production Analyst (cross-enterprise)
- Pricing Analyst requests a marketing proposal from Marketing Specialist (cross-organization)
- Sales Representative requests a price/restock change from Pricing Analyst (cross-organization)
- Store Manager requests an inventory analysis from Retails Analyst (cross-organization)
- Retail Analyst submits an order and tracks status with Wholesale Sales Rep (cross-enterprise)
- Store Manager and Store Associate exchange restock/count tasks (same organization)

Each work request is created with a sender, a message, and a status. It is placed into the receiving organization's queue as well as the sender's own queue, so both sides can track it. Status moves from Sent to Pending/Processing to Completed as it is handled.

Design and Implementation
The system will be built in Java using Swing for the user interface (NetBeans, JDK 19), following an object model of Network, Enterprise, Organization, Role, and Work Request. Each role will have its own work area screen for viewing and creating work requests.

Planned Features include:
Role-based login with username and password for each user account
A reporting module summarizing work requests and inventory data
A configuration module with pre-populated test data (using the Faker module) to demo the system
Form validation (required fields, valid prices, valid emails) and error handling
Data storage so information persists between runs
CRUD (Create/Read/Update/Delete) operations, including System Admin and Enterprise Admin account management."